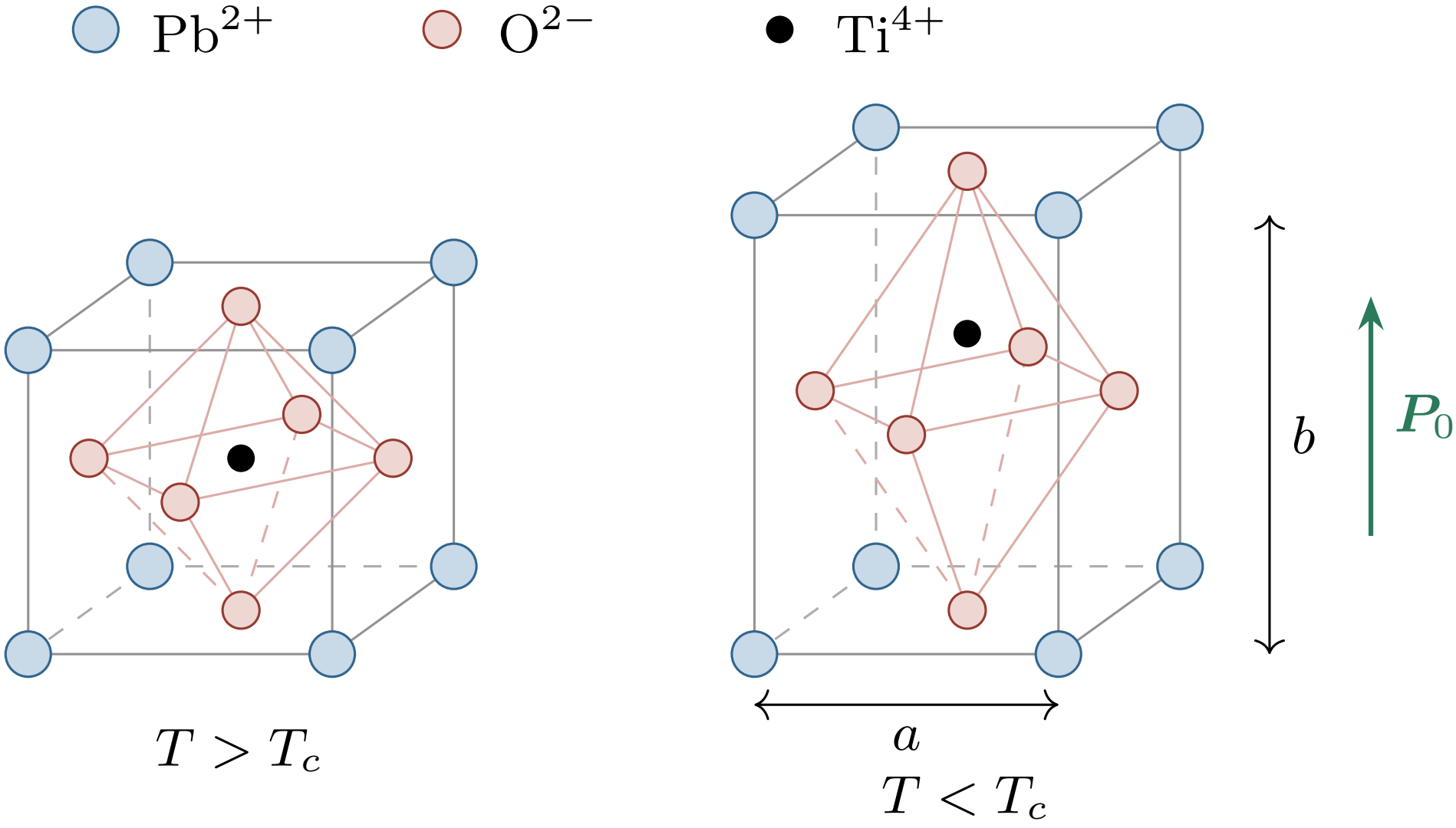
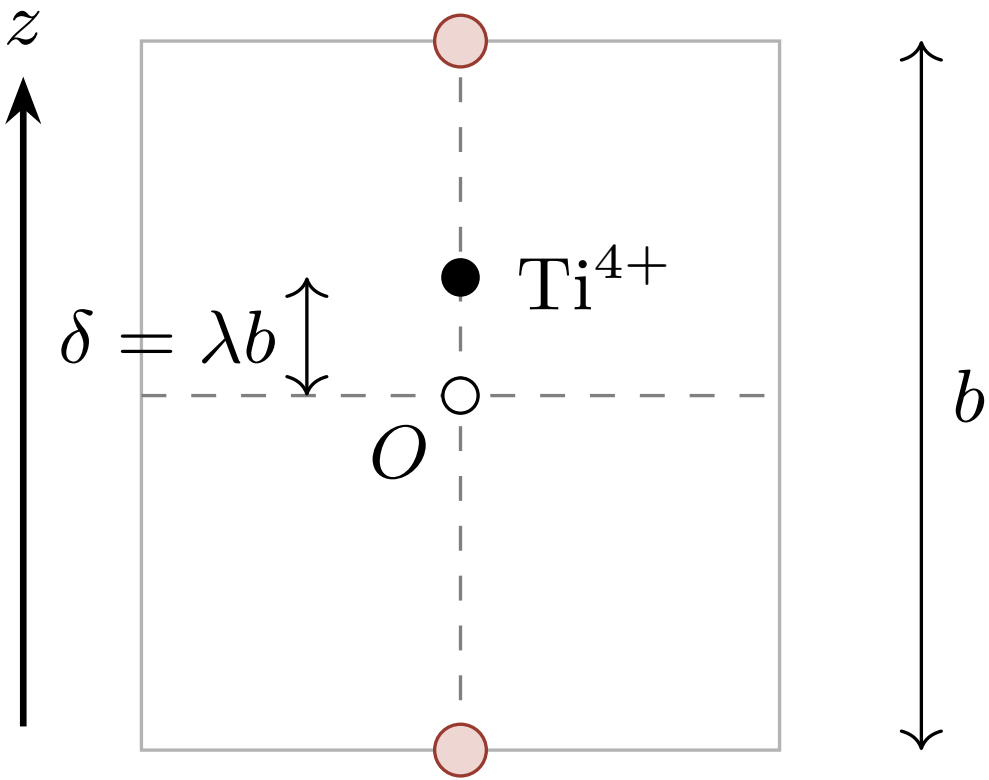
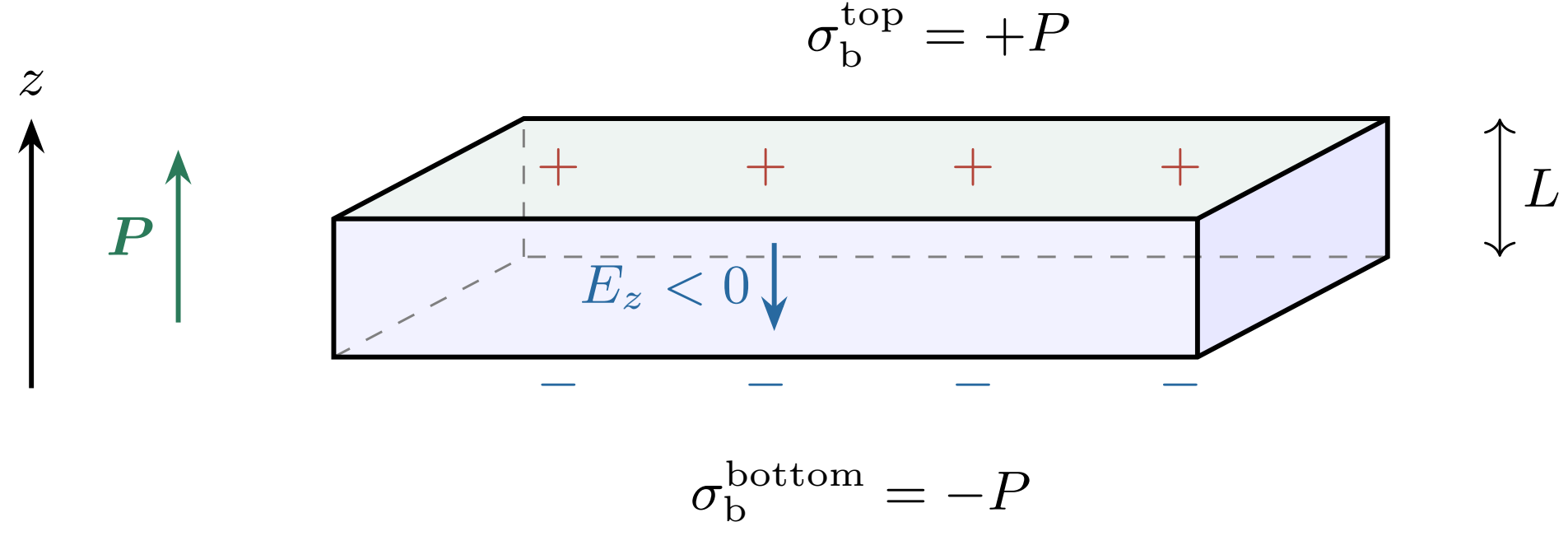
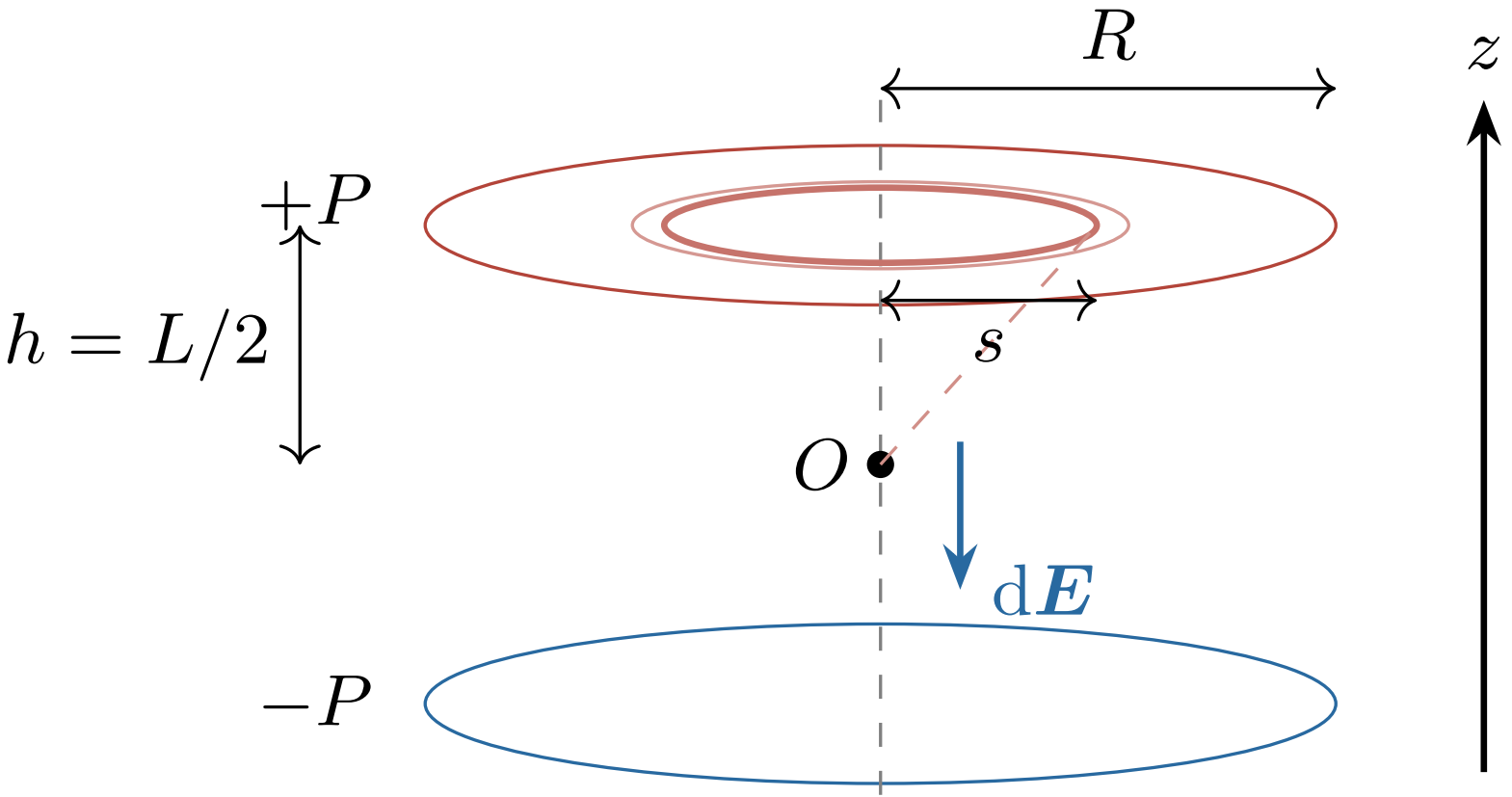

Pb^{2+}
O^{2-}
Ti^{4+}
$T > T_c$
b
P_0
a
$T < T_c$



对称背景的偶极矩为零

$$p_{\mathrm{cell},z} = 4e\delta$$


$\sigma_{\mathrm{b}}^{\mathrm{top}} = +P$
z
$\boldsymbol{P}$
$E_z < 0$
L
$\sigma_{\mathrm{b}}^{\mathrm{bottom}} = -P$


R
z
+P
h = L/2
s
O
dE
−P